

SET 9: ONLINE LEARNING ACTIVITY

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	K. Deeksha Sree
Register Number	192424097
Date	03 October 2026
Day	Saturday

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

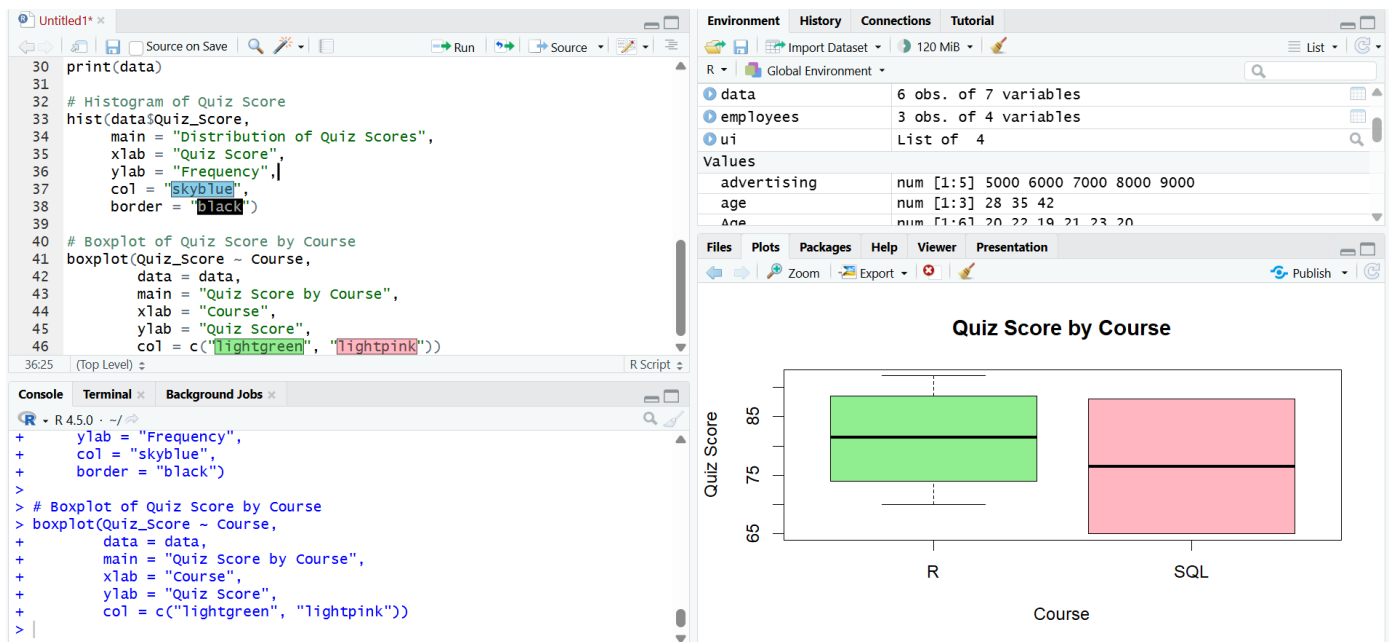
Program 1: Histogram and Boxplot of Quiz Score

Objective: Plot a histogram for Quiz_Score and a boxplot of Quiz_Score by Course. Add title and labels and interpret student performance.

```
data <- read.csv("Online_Learning_Activity.csv")
```

```
# Histogram of Quiz Score
hist(data$Quiz_Score,
     main = "Distribution of Quiz Scores",
     xlab = "Quiz Score",
     ylab = "Frequency",
     col = "skyblue",
     border = "white")
```

```
# Boxplot of Quiz Score by Course
boxplot(Quiz_Score ~ Course,
       data = data,
       main = "Quiz Score by Course",
       xlab = "Course",
       ylab = "Quiz Score",
       col = c("lightgreen", "lightpink"))
```



Observation: The histogram shows the distribution of quiz scores. The boxplot compares score spread by course. In this dataset, R-course scores are 78, 85, 92 and 70, while SQL-course scores are 65 and 88; the R scores are generally higher, while the SQL scores show a wider observed spread.

Program 2: Scatter Plot with Bubble Size and Transparency

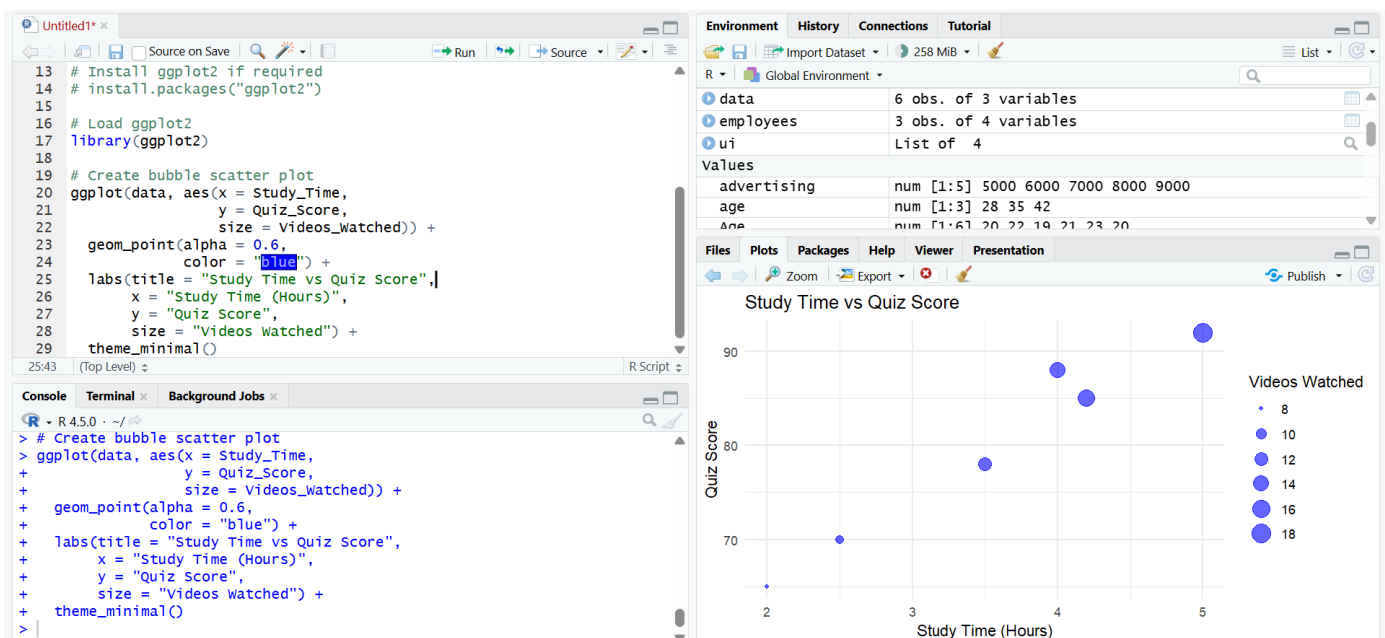
Objective: Plot Study_Time against Quiz_Score, represent Videos_Watched using bubble size, and apply transparency.

```
library(ggplot2)

data <- read.csv("Online_Learning_Activity.csv")

ggplot(data, aes(x = Study_Time..hrs.,
                  y = Quiz_Score,
                  size = Videos_Watched)) +
  geom_point(alpha = 0.6) +
  labs(title = "Study Time vs Quiz Score",
       x = "Study Time (hrs)",
       y = "Quiz Score",
       size = "Videos Watched") +
  theme_minimal()
```

Note: With read.csv(), the column name Study_Time (hrs) is normally converted to Study_Time..hrs. because R makes syntactically valid column names by default.



Observation: The scatter plot shows the relationship between study time and quiz score. Larger bubbles represent more videos watched. In this six-student dataset, higher study time generally corresponds to higher quiz scores, although the sample is small.

Program 3: Monthly Average Quiz Score and Moving Average

Objective: Convert Login_Date into Date format, calculate average quiz score per month, plot a line chart, and apply moving-average smoothing.

```
library(ggplot2)
library(dplyr)
library(zoo)

data <- read.csv("Online_Learning_Activity.csv")
data$Login_Date <- as.Date(data$Login_Date)
data$Month <- format(data$Login_Date, "%Y-%m")
```

```

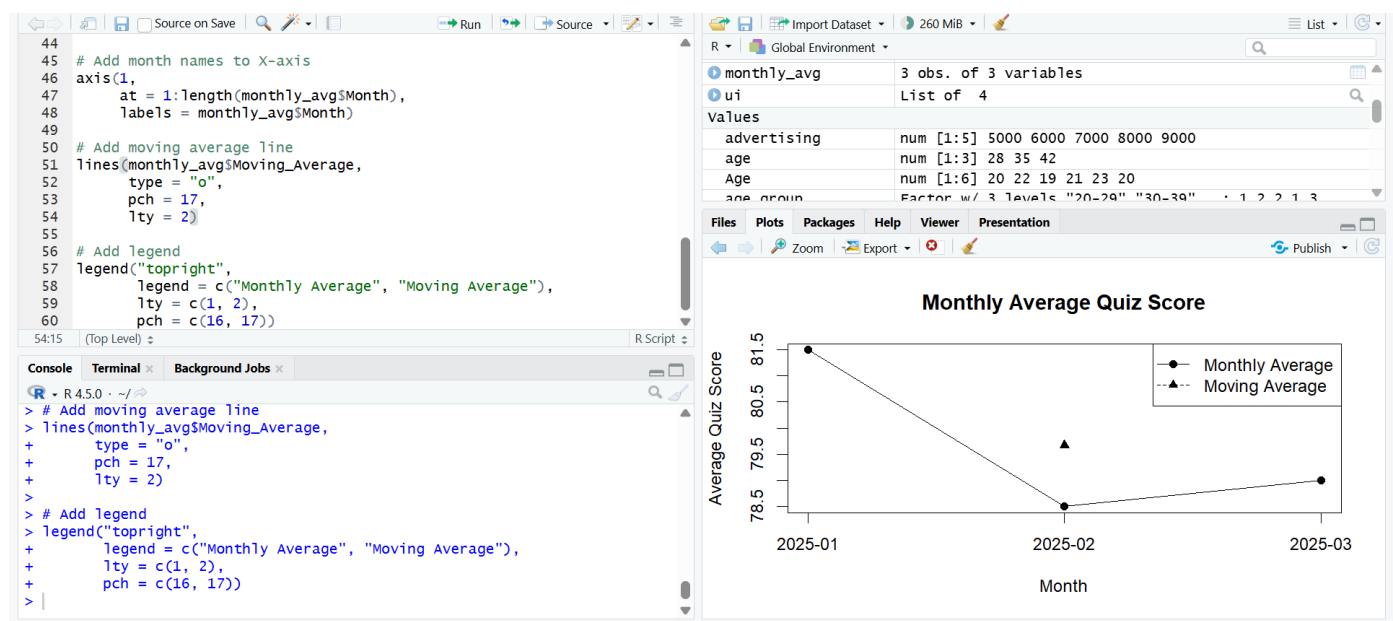
monthly_avg <- data %>%
  group_by(Month) %>%
  summarise(Average_Quiz_Score = mean(Quiz_Score))

monthly_avg$Moving_Average <- zoo::rollmean(
  monthly_avg$Average_Quiz_Score,
  k = 3,
  fill = NA,
  align = "center"
)

ggplot(monthly_avg, aes(x = Month, y = Average_Quiz_Score, group = 1)) +
  geom_line() +
  geom_point() +
  geom_line(aes(y = Moving_Average),
            linetype = "dashed", na.rm = TRUE) +
  labs(title = "Monthly Average Quiz Score with Moving Average",
       x = "Month",
       y = "Average Quiz Score") +
  theme_minimal()

```

Monthly averages: January 2025 = 81.5, February 2025 = 78.5, and March 2025 = 79.0. The 3-month centered moving average is approximately 79.67 for the middle month. The average decreases from January to February and then rises slightly in March.



Program 4: Tableau Mini Interactive Dashboard

Objective: Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score by Course, a scatter plot of Study Time vs Quiz Score, add Gender and Course filters, and design a mini interactive dashboard.

A. Import the CSV

- Open Tableau → Connect → Text File.
- Select Online_Learning_Activity.csv.
- Open a new worksheet after the data is loaded.

B. Bar Chart – Average Quiz Score by Course

- Create a new worksheet and rename it Average Quiz Score by Course.

- Drag Course → Columns.
- Drag Quiz_Score → Rows.
- Click Quiz_Score on Rows → Measure → Average.
- On the Marks card, select Bar.
- Use Worksheet → Show Title and set the title to Average Quiz Score by Course.

C. Scatter Plot – Study Time vs Quiz Score

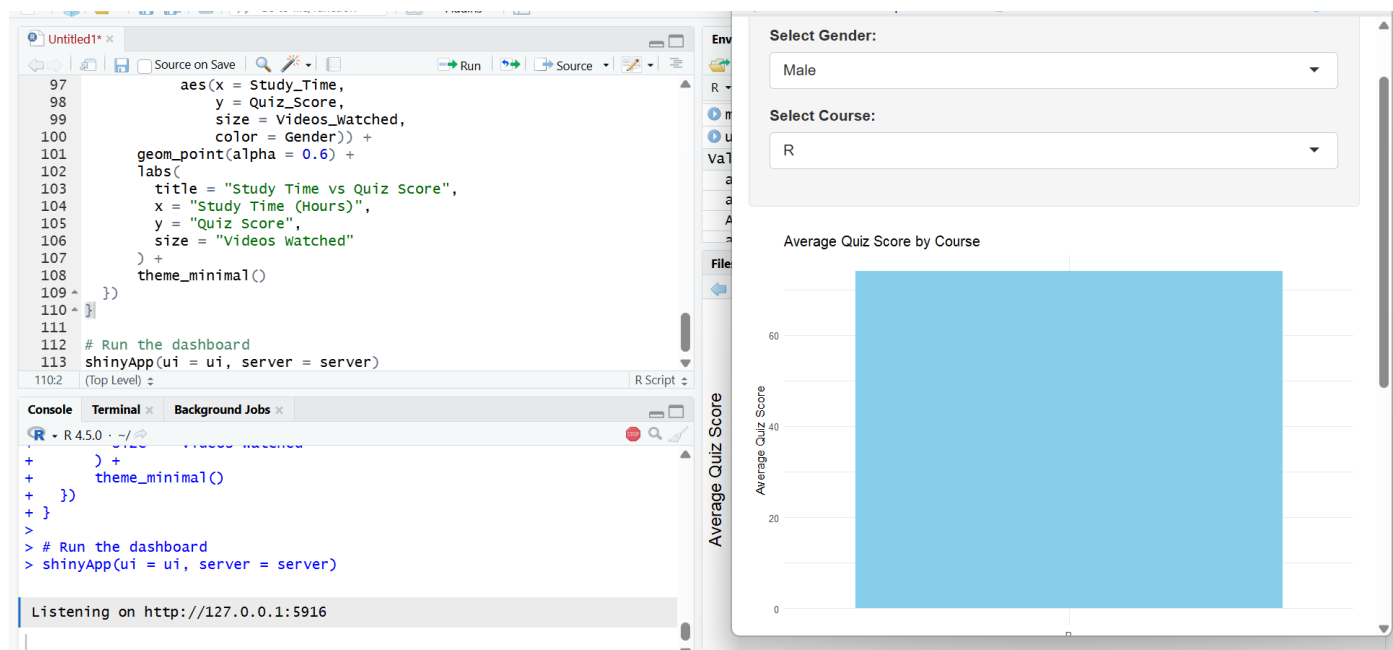
- Create another worksheet and rename it Study Time vs Quiz Score.
- Drag Study_Time (hrs) → Columns.
- Drag Quiz_Score → Rows.
- On the Marks card, choose Circle.
- Drag Student_ID → Detail so each student is a separate mark.
- Use Worksheet → Show Title and set the title to Study Time vs Quiz Score.

D. Add Gender and Course Filters

- Drag Gender → Filters and select all values initially.
- Drag Course → Filters and select all values initially.
- Right-click each filter and choose Show Filter.

E. Create the Mini Dashboard

- Go to Dashboard → New Dashboard.
- Drag Average Quiz Score by Course onto the dashboard.
- Drag Study Time vs Quiz Score onto the dashboard.
- Add the Gender and Course filters and set them to apply to the relevant worksheets.
- Use the filters to interactively explore quiz performance by gender and course.



Procedure

1. Import the Online_Learning_Activity.csv dataset into R.

2. Plot the Quiz_Score histogram and Quiz_Score-by-Course boxplot.
3. Create the Study_Time versus Quiz_Score bubble scatter plot with transparency.
4. Convert Login_Date to Date format, calculate monthly average quiz scores, and apply a 3-month moving average.
5. Import the CSV into Tableau.
6. Create the average quiz-score bar chart by Course.
7. Create the Study Time vs Quiz Score scatter plot.
8. Add Gender and Course filters.
9. Combine the visualizations into a mini interactive dashboard.

Result

The Online Learning Activity dataset was successfully visualized using R and Tableau. Histogram, boxplot, bubble scatter plot, monthly line chart with moving-average smoothing, and a Tableau mini interactive dashboard were created to study student learning activity and quiz performance.